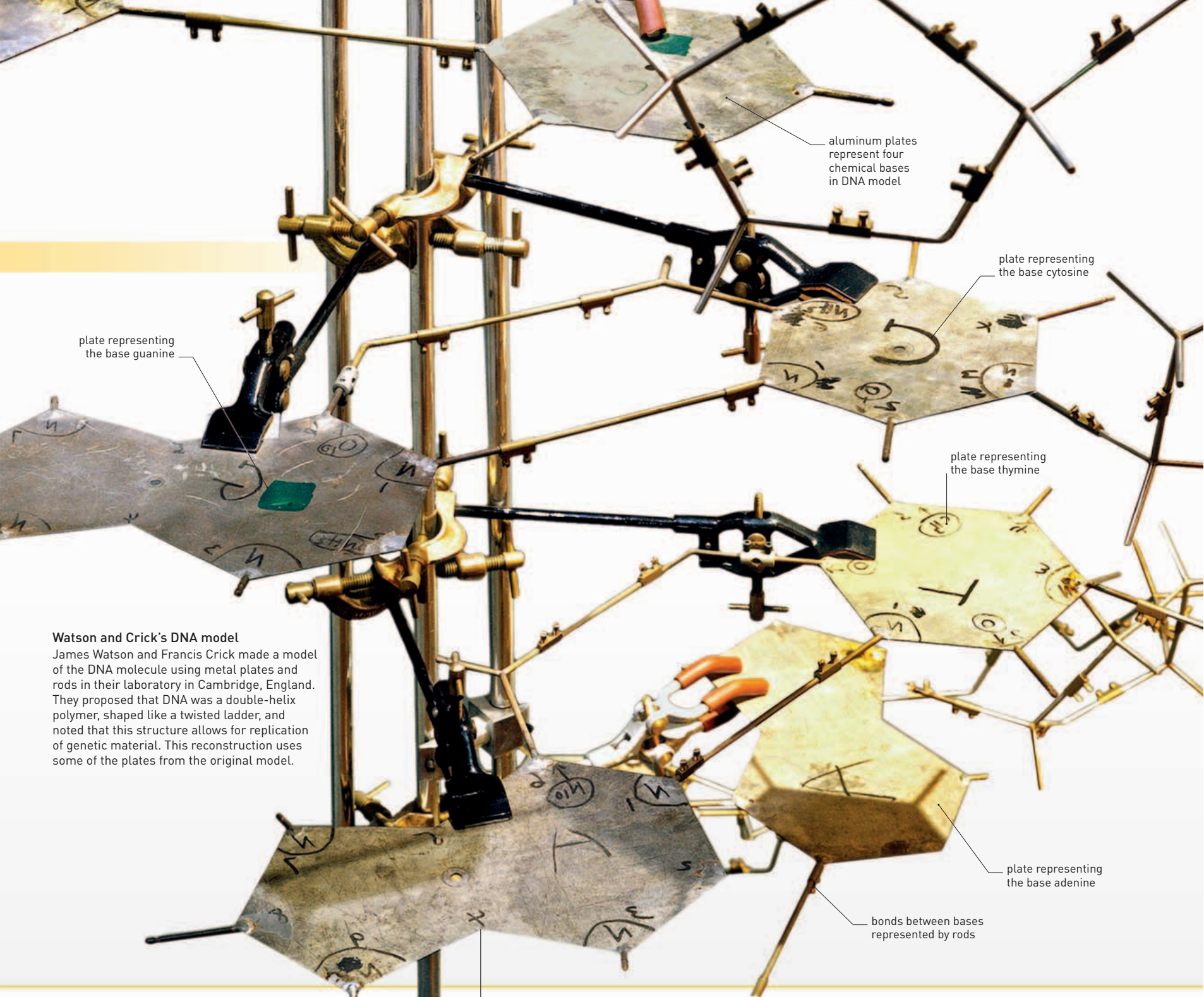




aluminum plates represent four chemical bases in DNA model

plate representing the base cytosine

plate representing the base guanine

plate representing the base thymine

plate representing the base adenine

bonds between bases represented by rods

Watson and Crick's DNA model

James Watson and Francis Crick made a model of the DNA molecule using metal plates and rods in their laboratory in Cambridge, England. They proposed that DNA was a double-helix polymer, shaped like a twisted ladder, and noted that this structure allows for replication of genetic material. This reconstruction uses some of the plates from the original model.

Early 20th century

Role of chromosomes in heredity
Working with fruit flies, American geneticist Thomas Hunt Morgan establishes that genes controlling heredity are positioned along chromosomes, and links the inheritance of a specific trait with a particular chromosome.

1953

Structure of DNA discovered
American biologist James Watson and British biologist Francis Crick discover that the DNA molecule consists of two helical chains of nucleotides wound loosely around each other.

1972

Recombinant DNA
American biochemists Paul Berg and Herb Boyer produce the first recombinant DNA molecules (recombinant DNA is DNA that has been created artificially), an achievement that is considered the birth of modern biotechnology.

1905

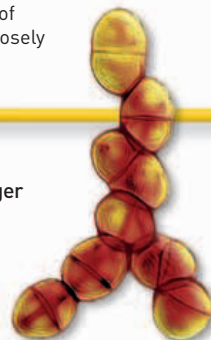
Sex chromosomes identified
American geneticists Nettie Stevens and E. B. Wilson independently identify the XY chromosome sex-determination system: males have XY and females have XX sex chromosomes.



X chromosome

1940-44

DNA identified as genetic messenger
Using *Streptococcus pneumoniae* bacteria, American scientists Oswald Avery, Colin MacLeod, and Maclyn McCarty discover that DNA is the hereditary material in most living organisms.



Streptococcus pneumoniae

1989-present

Human Genome Project
The Human Genome Organization maps the human DNA sequence and discovers it contains only about 20,000 to 25,000 genes. Full analysis of the results continues.